

THE ARITHMETIC CATEGORY AND STABLE RECOVERY OF LATTICE FACTORS

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ABSTRACT. We prove stable canonical recovery for scalar-twisted group factors of ICC groups commensurable with property- (T) lattices over characteristic-zero local fields. Every specified stable isomorphism to a scalar-twisted factor of any countably infinite ICC group recovers the group, the cocycle up to a normalized cochain, and equal amplification scales, with an implementing partial isometry in the given matrix corners. We compute the arithmetic correspondence subcategory for arbitrary countable ICC groups, including all bounded maps, summands, conjugates, and Connes fusion, and combine it with the companion's arithmetic exhaustion theorem. Multiplication also recovers arbitrary finite-index factor neighbors, with the exact projective condition on the matched cocycle ratio.

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1. INTRODUCTION

A group factor is built from the multiplication of a discrete group, but an isomorphism of factors need not preserve its distinguished group unitaries. We ask when the group, a scalar twist of its multiplication, and the given factor isomorphism can all be recovered. For the lattice class below, we prove this recovery even after amplification and with an arbitrary ICC comparison group.

Connes proved countability of the fundamental group of a property- (T) group factor [9]; his later problems ask whether nonisomorphic ICC property- (T) groups have nonisomorphic factors, and ask for the fundamental groups of those factors [10, Chapter V, Appendix B.ε, Problems 1–2]. The stronger stable formulation displayed by Popa asks for recovery of the given isomorphism up to a character and inner conjugacy, with amplification scale one [24, Section 3, Equations (3.2) and (3.2')].

The proof uses the more general relation of finite correspondence: two factors act normally and commute on one Hilbert space, with finite module dimensions on both sides. We determine these correspondences through isomorphisms between finite-index subgroups and finite-dimensional projective representations. The representation records the ratio of the two scalar cocycles on the matched subgroups. Neither cocycle separately need have a finite-dimensional realization.

Our algebraic calculation identifies every bounded map, summand, and composition of these arithmetic models for arbitrary countable ICC groups. The geometric exhaustion theorem of the companion [20, Theorem 1.2] makes the models exhaustive for the lattice class. Their integral dimensions recover stable isomorphisms; their compatibility with multiplication also recovers arbitrary finite-index factor neighbors.

1.1. The common conventions and lattice class.

Definition 1.1. Let $\mathcal{K}$ be the class of countably infinite ICC groups abstractly commensurable with a lattice in a finite nonempty product

$$G = \prod_{i=1}^r G_i, \quad G_i = \mathbf{H}_i(k_i)^+.$$

Here k_i is $\mathbb{R}$, $\mathbb{C}$, or a finite extension of $\mathbb{Q}_p$; the primes may differ. The algebraic group $\mathbf{H}_i$ is connected, adjoint, and absolutely simple over k_i . The superscript $+$ denotes the subgroup generated by the k_i -points of the unipotent radicals of proper k_i -parabolic subgroups. Each G_i is required to be noncompact and to have Kazhdan's property (T) .

A lattice is discrete and has finite Haar covolume. Abstract commensurability means an actual isomorphism between finite-index subgroups. ICC means that every nonidentity conjugacy class is infinite. No irreducibility, cocompactness, product decomposition, or torsion hypothesis is imposed on the lattice.

For example, $\mathrm{SL}_3(\mathbb{Z})$ and every finite-index subgroup belong to $\mathcal{K}$. Section 5 verifies the lattice and ICC hypotheses for these and other named examples.

Property (T) means that every continuous unitary representation with almost invariant unit vectors has a nonzero invariant vector. Standard structure and permanence results for this class are recalled in Section 2; see also [2].

All scalar cocycles take values in $\mathbb{T}$ and are normalized:

$$\mu(g, h)\mu(gh, j) = \mu(h, j)\mu(g, hj), \quad \mu(e, g) = \mu(g, e) = 1.$$

The twisted group factor $L_\mu(\Gamma)$ has canonical unitaries u_g with $u_g u_h = \mu(g, h)u_{gh}$ and trace $\tau(u_g) = \mathbf{1}_{\{e\}}(g)$. For an ICC group it is a factor; Section 2 recalls the short Fourier argument.

A bifinite M – N correspondence is a Hilbert space with commuting normal unital left M and right N actions and finite von Neumann module dimensions on both sides. The actions need not be mutual commutants, and the correspondence need not be irreducible.

Every correspondence in this paper is separable; every factor isomorphism is normal, unital, complex-linear, and $*$ -preserving. For a finite factor Q , write

$$Q^t = p(M_n(\mathbb{C}) \overline{\otimes} Q)p, \quad (\mathrm{Tr}_n \otimes \tau_Q)(p) = t > 0.$$

The matrix trace Tr_n is unnormalized; the corner itself has its normalized trace. The fundamental group of Q is $\mathcal{F}(Q) = \{t > 0 : Q^t \cong Q\}$.

1.2. Stable canonical recovery. A stable comparison is an isomorphism between positive amplifications of two factors. The following theorem forces the amplification scales to agree and determines the given map on its actual matrix corners.

Theorem 1.2 (Presentation equivalence and stable canonical recovery). *Let $\Lambda \in \mathcal{K}$ and let Π be any countably infinite ICC group. Let ω, ν be arbitrary normalized scalar cocycles on them, and let $t, s > 0$. Then*

$$L_\omega(\Lambda)^t \cong L_\nu(\Pi)^s$$

if and only if $t = s$ and there are an isomorphism $\delta : \Lambda \rightarrow \Pi$ and a normalized cochain $b : \Lambda \rightarrow \mathbb{T}$ such that

$$(1.1) \quad \omega(g, h) = b(g)b(h)b(gh)^{-1}\nu(\delta(g), \delta(h)).$$

Every specified isomorphism at unit scale has the form

$$\theta(u_g) = w b(g) v_{\delta(g)} w^*$$

for a unitary w in the target.

More explicitly, put $M = L_\omega(\Lambda)$ and $N = L_\nu(\Pi)$, and let

$$\theta : pM_n(M)p \longrightarrow qM_m(N)q$$

be a specified isomorphism, where the unnormalized traces of p and q are t and s . Then $t = s$, and one may choose δ, b as above and $z \in M_{m,n}(N)$ so that, for the canonical map $\varphi(u_g) = b(g)v_{\delta(g)}$,

$$z^*z = \varphi^{(n)}(p), \quad zz^* = q, \quad \theta(x) = z\varphi^{(n)}(x)z^*.$$

Whenever such an isomorphism exists, the recovered comparison group Π belongs to $\mathcal{K}$. In particular every factor $L_\omega(\Lambda)$ in this class has trivial fundamental group.

The untwisted specialization answers the stable recovery question for Definition 1.1, with an arbitrary ICC comparison group. Property (T) alone does not imply this recovery. OpenAI constructed a countably infinite family of pairwise nonisomorphic ICC property-(T) groups with isomorphic group factors [21, Chapter 4, Theorem 1.2]. Independently and concurrently, Zhou constructed another pair [27, Theorem A].

1.3. Arbitrary finite-index factor neighbors. We next allow the second factor to have no given group presentation. A *finite-index factor neighbor* of M is a factor joined to M by a nonzero bifinite correspondence. The arithmetic data describing such a neighbor involve finite-index subgroup isomorphisms rather than an isomorphism of the whole groups.

For a group A , a normalized projective unitary representation with multiplier β is a map $\sigma : A \rightarrow \mathcal{U}(V)$ satisfying $\sigma(a)\sigma(b) = \beta(a, b)\sigma(ab)$ and $\sigma(e) = I$. We call β *finite type* when such a representation exists with $0 < \dim_{\mathbb{C}} V < \infty$; its image need not be finite. For a group homomorphism δ , the pullback cocycle is $\delta^*\omega(a, b) = \omega(\delta(a), \delta(b))$.

Theorem 1.3 (All factor endpoints). *Let $\Gamma \in \mathcal{K}$, let μ be any normalized scalar cocycle on Γ , and let P be a II_1 factor with separable predual. There is a nonzero bifinite $L_\mu(\Gamma)$ – P correspondence if and only if*

$$P \cong L_\omega(\Lambda)^t$$

for a countably infinite ICC group Λ , a normalized scalar cocycle ω on Λ , and $t > 0$, for which there are finite-index subgroups $A \leq \Gamma$ and $B \leq \Lambda$, an actual isomorphism $\delta : A \rightarrow B$, and a nonzero finite-dimensional projective unitary representation of A with multiplier

$$\frac{\mu|_{A \times A}}{\delta^*(\omega|_{B \times B})}.$$

In particular, the reconstructed group Λ belongs to $\mathcal{K}$.

The unknown factor P is not assumed to have a group presentation or a preferred basis. Its presentation is recovered from multiplication in a finite-index extension. If $\mu = 1$, finite-index projective induction recovers the usual finite-type condition on ω . For arbitrary μ the ratio condition is essential: $P = L_\mu(\Gamma)$ is always an endpoint.

1.4. The arithmetic correspondence category. The two recovery theorems use a common classification of correspondences. We first describe its models and then the finite-dimensional data that control their maps and summands.

For finite-index subgroups $A \leq \Gamma$ and $B \leq \Lambda$, an isomorphism $\delta : A \rightarrow B$, and a nonzero finite-dimensional projective unitary representation σ satisfying

$$(1.2) \quad \sigma(a)\sigma(b) = \frac{\mu(a,b)}{\omega(\delta(a),\delta(b))}\sigma(ab), \quad \sigma(e) = I,$$

define

$$(1.3) \quad E(A, B, \delta, \sigma) = L^2(M) \otimes_{L_{\mu|_A}(A)} (V_\sigma \otimes L^2(N)), \quad M = L_\mu(\Gamma), \quad N = L_\omega(\Lambda).$$

The right N action is standard. The left subgroup action on the second factor is $u_a(\xi \otimes \eta) = \sigma(a)\xi \otimes v_{\delta(a)}\eta$. Equation (1.2) makes this an action with multiplier $\mu|_A$.

A right action is represented by $\rho(h)\xi = \xi v_h^*$, so $\rho(h)\rho(k) = \overline{\omega(h,k)}\rho(hk)$. The left action is $\pi(g)\xi = u_g\xi$. Together they give the joint action $U_{(g,h)}\xi = u_g\xi v_h^*$.

A *virtual-isomorphism germ* from Γ to Λ is an isomorphism between finite-index subgroups, modulo agreement on a common finite-index subgroup. The action of $\Gamma \times \Lambda$ on germs is

$$(g, h) \cdot [\delta] = [\text{Ad}(h) \circ \delta \circ \text{Ad}(g^{-1})].$$

For a germ d let C_d denote its full stabilizer. ICC implies that C_d is the graph of an isomorphism $A_d \rightarrow B_d$ between finite-index subgroups; the proof is included in Section 3. Use the joint multiplier

$$(1.4) \quad \alpha((g, h), (g', h')) = \mu(g, g')\overline{\omega(h, h')}.$$

For a finite-dimensional $\alpha|_{C_d}$ -projective representation σ_d , define

$$H(d, \sigma_d) = E(A_d, B_d, \delta_d, \tilde{\sigma}_d), \quad \tilde{\sigma}_d(a) = \sigma_d(a, \delta_d(a)).$$

Here $\delta_d : A_d \rightarrow B_d$ is the isomorphism whose graph is C_d . Section 3 realizes $H(d, \sigma_d)$ as the orthogonal sum of copies of the representation space indexed by $(\Gamma \times \Lambda)/C_d$, and describes the induced transports between them.

The full stabilizer matters because a graph presentation can split one of these fibres into several pieces. If $C_\delta = \{(a, \delta(a)) : a \in A\}$ represents d , the finite set C_d/C_δ indexes the pieces that must be grouped together. Their projective representation is induced from C_δ to C_d with multiplier α . Bimodule projections may mix these pieces; they become fibre projections only after this grouping.

Theorem 1.4 (All finite correspondences). *Let $\Gamma \in \mathcal{K}$, let Λ be any countably infinite ICC group, and let μ and ω be arbitrary normalized scalar cocycles on these groups.*

- (i) *Every nonzero bifinite $L_\mu(\Gamma)$ – $L_\omega(\Lambda)$ correspondence is a closed bimodule summand of a finite direct sum of the models in Equation (1.3).*
- (ii) *Every such correspondence has a finite orthogonal decomposition into full-germ orbit bundles. After choosing distinct germ orbits and allowing zero fibres, all bounded bimodule maps are*

$$(1.5) \quad \text{Hom}_{M-N} \left(\bigoplus_d H(d, \sigma_d), \bigoplus_d H(d, \tau_d) \right) = \bigoplus_d \text{Hom}_{C_{d,\alpha}}(\sigma_d, \tau_d).$$

Norm, adjoint, and composition are fibrewise. Projection summands are exactly projective-representation summands in the fibres. A correspondence $K = \bigoplus_d H(d, \sigma_d)$ has dimensions

$$\dim_M K = \sum_d \dim_{\mathbb{C}}(\sigma_d)[\Lambda : B_d], \quad \dim_N K = \sum_d \dim_{\mathbb{C}}(\sigma_d)[\Gamma : A_d].$$

Both are positive integers when $K \neq 0$.

- (iii) *Conjugation and Connes fusion correspond to conjugation and balanced tensor product of these bundles. They are equivalently computed by finite double-coset induction. The scalar transports are given explicitly in Equations (3.4) and (3.17) and Section 3.*

The integer dimensions in part (ii) are the dimensions of the actual correspondence, including a proper summand of a presenting model. That distinction is essential to the stable recovery argument: the two dimensions of an equivalence correspondence are reciprocal, so here both must equal one. The projective data throughout concern only the matched ratio in Equation (1.2); neither individual endpoint cocycle is assumed finite type.

Theorems 1.2 and 1.3 will follow from this map-level classification in Section 4. These are structural classifications; they do not assert a decision procedure for arbitrary group presentations or cocycle classes.

For untwisted factors, the same results identify the outer automorphism group and recover the group from its reduced C^* -algebra whenever that algebra has a unique tracial state. Section 5 also gives commensurability and finite-type neighbor criteria, named lattice examples, and the finite-index endomorphism consequence for $L(\text{SL}_3(\mathbb{Z}))$. The last consequence uses a separate classical covolume argument.

1.5. History and the role of the arithmetic category. Popa’s deformation/rigidity theory produced rigidity theorems for crossed products and Bernoulli actions. Ioana’s W^* -superrigidity theorem for Bernoulli actions of ICC property-(T) groups concerns an action together with its crossed product [14]. Ioana–Popa–Vaes obtained the first W^* -superrigid groups, in classes of generalized wreath products [15]. Chifan–Ioana–Osin–Sun constructed the first such groups with property (T), using wreath-like products [8]. Rigidity of an action and recovery from a bare group factor are distinct problems, and the correspondence category asks additionally for finite index, multiplicities, and composition.

The geometric route for lattices has its own antecedents. For lattices in $\text{Sp}(n, 1)/\{\pm 1\}$, $n \geq 2$, Cowling–Haagerup’s completely bounded approximation invariant already determines n from the group factor [11, Main theorem and Propositions 6.1–6.2]. Bekka classified the characters of $\text{SL}_n(\mathbb{Z})$ for $n \geq 3$, and Peterson proved a finite-factor representation dichotomy for irreducible higher-rank property-(T) lattices [1, 22]. Boutonnet–Houdayer developed stationary and noncommutative boundary methods and recovered character rigidity in that framework [6]. These results provide important context for the companion’s geometry. They do not by themselves identify all finite correspondences to arbitrary ICC groups. The companion proves the exhaustion input in the full mixed-local-field class stated here, with reducible products and arbitrary scalar twists.

Jones’s index and the expectation inequality of Pimsner–Popa provide the dimension and positivity framework [16, 23]. We use Connes’s correspondence category and relative tensor product [10, Chapter V, Appendix B]. Complete finite-index classifications and explicit fusion rules already occur in Vaes’s work on generalized Bernoulli crossed products [26, Theorem 2.2 and Section 6.2]. Theorem 2.2 allows scalar twists at both endpoints and a finite-dimensional projective representation of their matched ratio; Definition 6.8 and Theorem 6.9 supply projective induction and the cocycle-corrected double-coset rule. We adapt this calculus to graph bundles for twisted group factors. The full stabilizer of a virtual-isomorphism germ identifies every bounded intertwiner and every selecting projection, including projections that mix smaller presenting fibres.

Donvil–Vaes prove rigidity under arbitrary scalar twists and virtual isomorphism for certain left-right wreath products [12]. Their Proposition 6.5 reconstructs an intermediate factor from finite-dimensional homogeneous spaces and their actual products, and Theorem 6.6 applies this construction to arbitrary bifinite factor neighbors. The source cocycle is already arbitrary there, and the conclusion retains a finite-type ratio of the two cocycles. We adapt their homogeneous-product and minimal-corner argument from automorphism degrees to a grading by virtual automorphism germs of Γ ; finite-index domains and boundedness of the homogeneous vectors are treated explicitly. The companion supplies the distinct exhaustion theorem for the local-field lattice class. The finite-type ascent used for an untwisted source is the projective-induction statement of [12, Lemmas 6.7–6.8].

Chifan–Fernández Quero–Osin–Tan obtain stable canonical recovery for scalar twists of certain property- (T) wreath-like products when the source cocycle is trivial on the abelian base [7, Theorem F]. Their comparison group is arbitrary ICC. The unrestricted source cocycles and lattice class treated here enter through the companion’s distinct exhaustion theorem.

Outside the property- (T) setting, the nonisomorphic free groups $\mathbb{F}_2$ and $\mathbb{F}_3$ satisfy $L(\mathbb{F}_2) \cong L(\mathbb{F}_3)$ [19, Theorem 1.2]. These factors have fundamental group $\mathbb{R}_{>0}$ [19, Corollary 6.1], in contrast to the trivial fundamental groups in Theorem 1.2.

1.6. Proof strategy and navigation. There are three algebraic steps. First, the graph models admit orthogonal finite-dimensional fibres indexed by graph cosets. Fibres with the same virtual-isomorphism germ must be grouped together before passing to a summand. The full stabilizer of a germ has only one finite orbit on the germ space. Square summability of each column of a bounded intertwiner therefore forces that map to preserve each grouped fibre. This proves the Hom formula and makes the two dimensions sums of integer fibre ranks times subgroup indices.

Second, a bifinite correspondence with an unknown factor produces a finite-index extension of the known group factor. Arithmetic exhaustion grades its L^2 space by finite-dimensional germ fibres. The Pimsner–Popa inequality turns those vectors into bounded operators. Products can then be computed on a common subgroup domain, and they multiply the germ degrees. In a minimal corner of the identity fibre, every nonzero degree is a unitary line. Those lines, with their actual products and trace, recover the group and scalar cocycle. The identity block supplies the projective representation of the cocycle ratio.

Third, the equivalence correspondence of a specified stable isomorphism has reciprocal dimensions. Their integrality forces both to be one, leaving one full graph and one one-dimensional projective fibre. The correspondence identification recovers the cochain. The column realizations in the given matrix corners then recover the implementing partial isometry for the specified map. Figure 1 separates these two consequences of the category calculation.

Section 2 records the relative tensor convention and exact companion interface. Section 3 proves all model formulas, then applies exhaustion. Section 4 reconstructs factor neighbors and specified stable maps. Section 5 proves the further consequences described above.

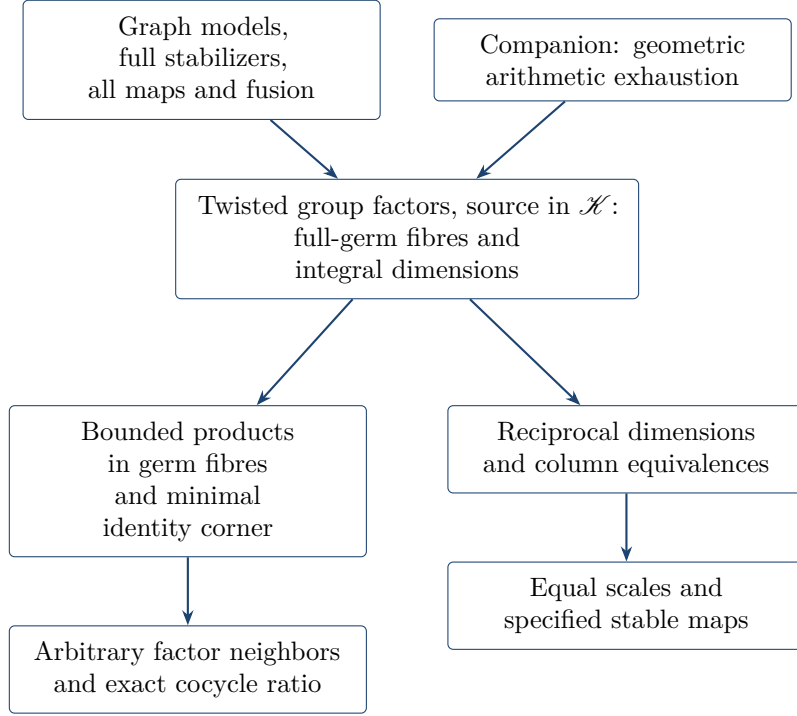


FIGURE 1. Arithmetic exhaustion from the companion supplies the geometric input for the main classification. The model calculus is independent of that input. Actual multiplication and reciprocal dimensions are separate final arguments.

2. FINITE MODULES AND THE EXHAUSTION INTERFACE

Our algebraic calculations need only countability and ICC. The geometric class enters at one precise point: the exhaustion statement below says that the explicit models contain every bifinite correspondence. We first fix the norm convention for their construction.

For completeness, $L_\mu(\Gamma)$ is a factor when Γ is ICC: the absolute values of the Fourier coefficients of a central L^2 vector are constant on conjugacy classes, so square summability kills every nonidentity coefficient.

2.1. Relative tensor products and elementary dimensions. All Hilbert-space inner products are linear in the second variable. A vector ξ in an M – N correspondence is *right bounded* if $R_\xi \hat{a} = \xi a$ extends to a bounded map from $L^2(N)$. For an N – P correspondence, the relative tensor product is the completion, after quotienting out zero-norm vectors, of the tensors with right-bounded first vector and inner product

$$(2.1) \quad \langle \xi \otimes_N \eta, \xi' \otimes_N \eta' \rangle = \langle \eta, (R_\xi^* R_{\xi'}) \eta' \rangle.$$

The right-module map $R_\xi^* R_{\xi'}$ is left multiplication by an element of N . Balancing means $(\xi a) \otimes_N \eta = \xi \otimes_N (a\eta)$ for $a \in N$. This is the finite tracial form of Connes’s relative tensor product [10, Chapter V, Appendix B, Theorems 13 and 20].

A finite right N -module has a column realization $pL^2(N)^n$, with dimension $(\text{Tr}_n \otimes \tau_N)(p)$. The vectors in pN^n are right bounded and dense. The projection model and its dimension are independent of the chosen realization [17, Theorem 10.1.1 and Proposition 10.1.3]. We prove the normality of the graph models directly in Lemmas 3.2 and 3.3. The same elementary construction precedes the geometric argument in the companion, so its exhaustion proof has no dependence on our all-maps or summand classification. For $n = \dim \sigma$, the elementary dimensions are $n[\Lambda : B]$ and $n[\Gamma : A]$ in left/right order. The dimensions of *actual summands* will require the full-stabilizer calculation.

2.2. The geometric input. The real simple factors in Definition 1.1 have higher rank or are locally $\mathrm{Sp}(n, 1)$, $n \geq 2$, or $F_{4(-20)}$. At a finite place their relative rank is at least two. Complex places are treated as real Lie groups in differential arguments. Property (T) passes between a locally compact group and a lattice and between a discrete group and a finite-index subgroup; see [2]. In particular, that class is closed under abstract commensurability among countably infinite ICC groups. The geometric companion treats arbitrary lattices in the permitted finite products, without a dense-projection hypothesis. We use exactly the following theorem.

Proposition 2.1 (Arithmetic exhaustion, companion Theorem 1.2). *Let $\Gamma \in \mathcal{K}$, let Λ be any countably infinite ICC group, and let μ, ω be arbitrary normalized scalar cocycles. Every nonzero separable bifinite $L_\mu(\Gamma)$ – $L_\omega(\Lambda)$ correspondence is a closed bimodule summand of a finite direct sum of the models $E(A, B, \delta, \sigma)$ in Equation (1.3), with the exact multiplier in Equation (1.2).*

This is the Arithmetic exhaustion theorem of the complete companion [20, Theorem 1.2]. Its conclusion retains actual finite-index subgroup isomorphisms and their exact scalar ratios. We do not assume that the projective representations have finite image, that either cocycle is finite type, or that the original correspondence is irreducible. The algebraic proofs that follow are independent of this proposition until its explicit application at the end of Section 3.

3. FULL-GERM CORRESPONDENCE CALCULUS

The calculations in this section are algebraic. We give a group-factor version of the projective induction and cocycle-corrected double-coset calculus in Vaes’s work on generalized Bernoulli crossed products [26, Theorem 2.2 and Section 6.2, especially Definition 6.8 and Theorem 6.9]. We prove the formulas here, including the full-stabilizer argument for every bounded intertwiner and every bimodule summand. Until we apply Proposition 2.1, the groups need only be countable and ICC; the geometric hypotheses of Definition 1.1 are not used. We first construct the arithmetic models with their exact scalar transports, and then determine their intertwiners, summands, conjugates, and relative tensor products. This order also verifies that every model appearing in the exhaustion is a normal bifinite correspondence.

Write

$$M = L_\mu(\Gamma), \quad N = L_\omega(\Lambda), \quad D = \Gamma \times \Lambda, \quad \alpha((g, h), (g', h')) = \mu(g, g') \overline{\omega(h, h')}.$$

On an M – N correspondence the joint group action is

$$(3.1) \quad U_{(g, h)} \xi = u_g \xi v_h^*.$$

It has multiplier α . Indeed, its rightmost product is $v_{h'}^* v_h^* = (v_h v_{h'})^*$, so $U_{(g, h)} U_{(g', h')} = \mu(g, g') \overline{\omega(h, h')} U_{(gg', hh')}$. In particular the two coordinate actions commute exactly.

3.1. Germs and their full stabilizers. A germ from Γ to Λ is an isomorphism $\delta : A \rightarrow B$ between finite-index subgroups, modulo agreement after restriction to a common finite-index subgroup. Composition means composition on a finite-index domain where both representatives are defined. For example, if $\varepsilon : C \rightarrow F$ is a representative from Λ to a third group, then $\varepsilon\delta$ is defined on $\delta^{-1}(B \cap C)$ and has image $\varepsilon(B \cap C)$. Both have finite index in the relevant endpoint group. Restricting representatives further does not change the germ, and a common further restriction verifies associativity. Inversion is represented by the inverse subgroup isomorphism.

For a germ d from Γ to Λ , let

$$(g, h)d = \mathrm{Ad}(h) d \mathrm{Ad}(g^{-1}), \quad C_d = \{(g, h) \in D : (g, h)d = d\}.$$

If $\delta : A \rightarrow B$ represents d , denote its graph by $C_\delta = \{(a, \delta(a)) : a \in A\}$. We distinguish this graph from the full stabilizer C_d .

Lemma 3.1 (Full stabilizers and the finite-orbit test). *The maps $g \mapsto [\text{Ad}(g)]$ and $h \mapsto [\text{Ad}(h)]$ are injective. The group C_d is the graph of an isomorphism $\delta_d : A_d \rightarrow B_d$ between finite-index subgroups. It contains C_δ at finite index, and δ_d extends δ . If a finite-index subgroup of C_d fixes another germ x from Γ to Λ , then $x = d$. Equivalently, the only finite C_d -orbit in the space of these germs is $\{d\}$. The same conclusion holds with C_d replaced by C_δ .*

Proof. The centralizer of a finite-index subgroup of an ICC group is trivial: an element of that centralizer has a centralizer of finite index and therefore a finite conjugacy class. This proves injectivity of the inner-germ maps.

The stabilizer condition can be written

$$(3.2) \quad \text{Ad}(h) = d \text{Ad}(g) d^{-1}.$$

If either coordinate of an element of C_d is the identity, injectivity of the inner-germ maps forces the other coordinate to be the identity. Thus both coordinate projections on C_d are injective. Their images A_d and B_d form the domain and range of a group isomorphism δ_d . The inclusion $C_\delta \subset C_d$ follows by evaluating Equation (3.2) on a common finite-index domain. Consequently $A \subset A_d$, $B \subset B_d$, and $\delta_d|_A = \delta$. Both projected groups have finite index, and $[C_d : C_\delta] = [A_d : A] < \infty$.

Suppose $C_0 \leq C_d$ has finite index and fixes x . Its first projection A_0 has finite index in Γ . For every $(a, b) \in C_0$ the two stabilizer equations give

$$(3.3) \quad x \text{Ad}(a) x^{-1} = \text{Ad}(b) = d \text{Ad}(a) d^{-1}.$$

Choose representatives χ and δ for x and d , and restrict a to $A_0 \cap \text{dom}(\chi) \cap \text{dom}(\delta)$. On this finite-index subgroup the outer expressions in Equation (3.3) are the inner germs of $\chi(a)$ and $\delta(a)$. Injectivity in Λ gives $\chi(a) = \delta(a)$, whence $x = d$. A finite orbit has a finite-index point stabilizer, so this proves the equivalent orbit statement. Finally, a finite-index subgroup of C_δ also has finite index in C_d . $\square$

The finite-orbit conclusion will make germ labels intrinsic to the induced models. Cosets of C_d correspond to actual germs, whereas several cosets of a presenting graph may represent the same germ. After grouping those cosets, a bounded intertwiner will have matrix blocks between the resulting finite-dimensional fibres. Equivariance makes a block's norm constant along a stabilizer orbit. The lemma will therefore force such a map to preserve each germ label, once square summability of its columns is used.

3.2. Projective induction and the arithmetic model. Projective induction, induction in stages, and double-coset tensor decompositions belong to Mackey's representation-theoretic framework [18, Theorems 4.1, 4.2, and 4.6]. The following coordinates fix the multiplier throughout and prove every scalar identity used here.

Let $C \leq D$ be a subgroup and let γ be a normalized unitary projective representation on a finite-dimensional space V , with multiplier $\alpha|_{C \times C}$. Write $X = D/C$ for left cosets. Choose representatives $r_x \in D$, with $r_C = e$, and give each $x \in X$ a copy V_x of V . Set $\mathcal{H}(C, \gamma) = \bigoplus_{x \in X} V_x$, with the counting Hilbert norm. For $s \in D$ put

$$(3.4) \quad \begin{aligned} c(s, x) &= r_{sx}^{-1} s r_x, \\ U_s^X(x) &= \frac{\alpha(s, r_x)}{\alpha(r_{sx}, c(s, x))} \gamma(c(s, x)) : V_x \longrightarrow V_{sx}. \end{aligned}$$

Lemma 3.2 (Exact projective induction). *Equation (3.4) defines a unitary projective representation of D with multiplier α . Changing the coset representatives gives a unitarily equivalent bundle by unitary identifications of its individual fibres. If C is the graph of a finite-index subgroup isomorphism $A \rightarrow B$, the bundle has commuting normal M and right N actions, and is bifinite.*

Proof. The correction $c(s, x)$ belongs to C , and

$$c(s, tx)c(t, x) = c(st, x).$$

Using the cocycle identity for α and the projective multiplication of γ in Equation (3.4) gives

$$(3.5) \quad U_s^X(tx)U_t^X(x) = \alpha(s, t)U_{st}^X(x).$$

One can also check all the scalar factors at once in the central extension with multiplication $(z, s)(w, t) = (zw\alpha(s, t), st)$: the representation of its preimage of C is $(z, c) \mapsto z\gamma(c)$, and the equality $sr_x = r_{sx}c(s, x)$ gives exactly the ratio in Equation (3.4). Thus this argument retains, rather than discards, the fixed multiplier.

For explicit independence of representatives, suppose $r'_x = r_x k_x$, where $k_x \in C$. The identification from the new coordinate copy of V_x to the old one is

$$F_x = \alpha(r_x, k_x)^{-1}\gamma(k_x).$$

Indeed identifying a fibre by $U_{r'_x}$ instead of U_{r_x} has precisely this effect. Equation (3.5) then gives $F_{sx}U_s'^X(x) = U_s^X(x)F_x$.

Now assume that C is a graph. Both coordinate groups act freely on X . For example, if $(g, e)r_x C = r_x C$, then $r_x^{-1}(g, e)r_x \in C$ has second coordinate e , so $g = e$. On an orbit of the first coordinate group choose a fibre V_{x_0} . The unitary identification

$$\ell^2(\Gamma) \otimes V_{x_0} \longrightarrow \bigoplus_{g \in \Gamma} V_{(g, e)x_0}, \quad \delta_g \otimes \xi \longmapsto U_{(g, e)}^X \xi$$

intertwines its action with $\lambda_\mu \otimes 1$, since $U_{(a, e)}U_{(g, e)} = \mu(a, g)U_{(ag, e)}$. Hence it extends normally to $L_\mu(\Gamma)$. The same argument for the second coordinate gives copies of $\lambda_{\bar{\omega}}$, which are the standard normal representations of the generators acting by right adjoints $\xi \mapsto \xi v_h^*$. The two actions commute, because the product cocycle has trivial mixed-coordinate factors.

The first coordinate orbits are indexed by Λ/B , and the second coordinate orbits by Γ/A . They are finite in number. Each is a regular module with multiplicity $\dim_{\mathbb{C}} V$, proving bifiniteness as well as the dimension formulas below. $\square$

We next identify this construction with the Connes-product model in Theorem 1.4. This also supplies a direct normality check of the subgroup action used to define that product.

Lemma 3.3 (The graph-bundle unitary). *Let $\delta : A \rightarrow B$ be a finite-index subgroup isomorphism and let $\sigma : A \rightarrow \mathcal{U}(V)$ have multiplier $\beta = \mu|_{A \times A}/\delta^*\omega$. Define $\gamma(a, \delta(a)) = \sigma(a)$. Then*

$$E(A, B, \delta, \sigma) = L^2(M) \bar{\otimes}_{L_{\mu|A}(A)} (V \otimes L^2(N)) \cong \mathcal{H}(C_\delta, \gamma).$$

The unitary sends the vector ξ in the fibre labelled by x to $U_{r_x} j\xi$, where $j\xi = \hat{1} \otimes (\xi \otimes \hat{1})$ in the displayed model.

Proof. On $V \otimes L^2(N)$ the subgroup action is

$$u_a(\xi \otimes \eta) = \sigma(a)\xi \otimes v_{\delta(a)}\eta.$$

Its multiplier is $\beta(a, b)\omega(\delta(a), \delta(b)) = \mu(a, b)$. For normality, choose representatives t for $B \setminus \Lambda$. On the summand spanned by $\xi \otimes v_{\delta(a)}v_t$, the map

$$(3.6) \quad \xi \otimes v_{\delta(a)}v_t \longmapsto \delta_a \otimes \sigma(a)^*\xi \quad \text{in } \ell^2(A) \otimes V$$

is unitary and conjugates the subgroup action to $\lambda_{\mu|A} \otimes 1$. To check the phase, applying u_b before this map produces

$$\omega(\delta(b), \delta(a))\sigma(ba)^*\sigma(b) = \omega(\delta(b), \delta(a))\beta(b, a)\sigma(a)^* = \mu(b, a)\sigma(a)^*.$$

Thus the left subgroup action is normal. Its right N action is the standard action on the second factor and commutes with it.

Choose representatives r for Γ/A . As a right $L_{\mu|A}(A)$ module, $L^2(M)$ has orthonormal free basis u_r ; the assertion follows directly from $E_{L_{\mu|A}(A)}(u_r^* u_{r'}) = \delta_{r,r'} 1$. The relative product therefore identifies with the orthogonal sum

$$(3.7) \quad E(A, B, \delta, \sigma) = \bigoplus_{r \in \Gamma/A} u_r \otimes (V \otimes L^2(N)).$$

At its identity fibre the graph action is exactly

$$U_{(a, \delta(a))} j \xi = u_a \otimes (\xi \otimes v_{\delta(a)}^*) = j \sigma(a) \xi.$$

For $g = ra$, balancing in Equation (3.7) gives the more explicit formula

$$U_{(g, h)} j \xi = \mu(r, a)^{-1} u_r \otimes (\sigma(a) \xi \otimes v_{\delta(a)}^* v_h^*).$$

The last Fourier coordinate is $\delta(a)h^{-1}$, up to its displayed unit scalar. Thus two translated identity fibres are orthogonal unless their labels in D/C_δ agree: equality of their first cosets and last Fourier coordinates is exactly that coset equality. Conversely these fibres exhaust every summand in Equation (3.7). They are an orthogonal, transitive system with graph action γ at the basepoint. Finally

$$U_s U_{r_x} j = \frac{\alpha(s, r_x)}{\alpha(r_{sx}, c(s, x))} U_{r_{sx} j \gamma(c(s, x))},$$

which proves that the stated unitary has exactly the transport of Equation (3.4). $\square$

For a full stabilizer we write

$$H(d, \gamma) = \mathcal{H}(C_d, \gamma).$$

Its labels identify with actual germs in the orbit Dd . There is also a canonical regrouping of an original graph model over this full-germ orbit. The map $D/C_\delta \rightarrow D/C_d$ has finite fibres. Its fibre above C_d carries

$$(3.8) \quad \text{Ind}_{C_\delta}^{C_d, \alpha} \gamma,$$

where projective induction uses Equation (3.4) with D replaced by C_d and with the restricted cocycle. Indeed that fibre is the sum indexed by C_d/C_δ , and its transports are precisely that formula. Regrouping the orthogonal sum consequently gives

$$\mathcal{H}(C_\delta, \gamma) \cong H(d, \text{Ind}_{C_\delta}^{C_d, \alpha} \gamma).$$

This description includes the phases of finite induction; it does not replace the projective representation by an ordinary one.

Changing an orbit representative also retains a scalar transport. If $d' = kd$, then $C_{d'} = kC_d k^{-1}$. Identify the new base fibre with the old one using U_k . For $c' = kck^{-1} \in C_{d'}$ the representation in these coordinates is

$$(3.9) \quad \gamma'(c') = \frac{\alpha(c', k)}{\alpha(k, c)} \gamma(c).$$

Indeed $U_{c'} U_k = \alpha(c', k) \alpha(k, c)^{-1} U_k U_c$. This gives the exact representative-change equivalence, including its restricted multiplier. Two choices of k leading to the same germ give equivalent fibres by the stabilizer action itself.

Proposition 3.4 (Dimensions of the actual orbit modules). *For a finite-dimensional $\alpha|_{C_d}$ -projective representation γ ,*

$$(3.10) \quad \dim_M H(d, \gamma) = \dim_{\mathbb{C}}(\gamma)[\Lambda : B_d], \quad \dim H(d, \gamma)_N = \dim_{\mathbb{C}}(\gamma)[\Gamma : A_d].$$

For the original model these formulas read

$$\dim_M E(A, B, \delta, \sigma) = \dim_{\mathbb{C}}(\sigma)[\Lambda : B], \quad \dim E(A, B, \delta, \sigma)_N = \dim_{\mathbb{C}}(\sigma)[\Gamma : A].$$

Zero fibres may be omitted. Dimensions add under finite orthogonal sums.

Proof. The free coordinate-orbit identifications in Lemma 3.2 exhibit exactly the stated numbers of copies of the standard left and right modules. Each standard module has dimension one for the normalized factor trace. Lemma 3.3 gives the original-model formulas. $\square$

3.3. All bounded intertwiners and all summands.

Theorem 3.5 (Full-germ intertwiner formula). *Choose finitely many distinct D -orbits of germs and a representative d in each orbit. Let γ_d, τ_d be finite-dimensional projective representations of C_d , with multiplier $\alpha|_{C_d}$; either may be zero. Then*

$$(3.11) \quad \text{Hom}_{M-N} \left(\bigoplus_d H(d, \gamma_d), \bigoplus_d H(d, \tau_d) \right) = \bigoplus_d \text{Hom}_{C_d, \alpha}(\gamma_d, \tau_d).$$

The equality is implemented by restriction to the representative fibres and extension by transport. Norm is the maximum of the fibre norms, and adjoints and composition are fibrewise. In particular all closed bimodule summands are obtained by taking invariant subspaces of these finite-dimensional fibre representations.

Proof. Realize both modules over their union of full-germ orbits, assigning a zero fibre where necessary. An intertwiner T has finite-dimensional matrix blocks $T_{yx} : V_x \rightarrow W_y$. If (e_i) is an orthonormal basis of V_x , then

$$(3.12) \quad \sum_y \|T_{yx}\|_{\text{HS}}^2 = \sum_i \|Te_i\|^2 \leq \dim(V_x) \|T\|^2 < \infty.$$

Equivariance identifies $T_{sy, sx}$ with $U_s^\tau(y) T_{yx} U_s^\gamma(x)^*$. Its Hilbert–Schmidt norm is therefore unchanged. For $s \in C_x$, Equation (3.12) implies that a nonzero block T_{yx} can occur only when the C_x -orbit of y is finite. By Lemma 3.1, this forces $y = x$. Thus every intertwiner is diagonal in the germ labels, including between initially different orbit bundles.

At a representative label d , equivariance is exactly the requirement that T_d intertwine the two $\alpha|_{C_d}$ -projective representations. Conversely, such an operator extends by

$$T_{sd} = U_s^\tau(d) T_d U_s^\gamma(d)^*.$$

This is independent of the choice of s : changing s by an element of C_d uses the intertwining identity, and the identical scalar multiplier on both sides cancels. The extension is bounded with norm $\|T_d\|$ on that orbit and commutes with every joint group generator. Normality of the endpoint representations then makes it a bimodule intertwiner. This proves Equation (3.11) and the norm assertion. The formulas for adjoint and composition follow on each fibre.

A closed bimodule summand is the range of a projection in the bounded bimodule commutant. The preceding description identifies that projection with a family of C_d -equivariant fibre projections. Their ranges are precisely the asserted projective-representation summands, and their transported ranges give the entire module summand. $\square$

Consider any closed bimodule summand K of a finite sum of the original graph models. Regroup each model by Equation (3.8), combine copies of the same full-germ orbit, and apply Theorem 3.5 to the projection onto K . We obtain a finite orthogonal sum of full-germ bundles, with exactly the intertwiners in Equation (3.11). In particular, if its fibre representations are ρ_d , its actual dimensions are

$$(3.13) \quad \dim_M K = \sum_d \dim_{\mathbb{C}}(\rho_d)[\Lambda : B_d], \quad \dim_N K = \sum_d \dim_{\mathbb{C}}(\rho_d)[\Gamma : A_d].$$

Both are positive integers for $K \neq 0$. This conclusion uses the description of its projection; the dimensions of a containing model alone would not imply it.

Remark 3.6 (Why a presenting graph must be enlarged). Let $M = N = L_\mu(\Gamma)$, with arbitrary μ , and suppose $A < \Gamma$ has finite index $k > 1$. Present the identity germ d by $\text{id}_A : A \rightarrow A$, with the one-dimensional trivial fibre. The multiplier on every diagonal pair

is $\mu(a, b)\overline{\mu(a, b)} = 1$, so this is a model for any source twist. Its full stabilizer is the diagonal copy of Γ . Regrouping therefore replaces the presenting fibre by the permutation representation on $\ell^2(\Gamma/A)$.

The constant vectors form an invariant line. Its orthogonal projection has every matrix entry equal to $1/k$, and selects $H(d, 1) = L^2(M)$ from the model $E(A, A, \text{id}_A, 1)$. The selected module has dimensions $(1, 1)$, whereas the containing model has dimensions (k, k) . On each of the k original one-dimensional fibres above d , the diagonal block of the selecting projection is $1/k$, which is not a projection. Grouping those fibres at the full stabilizer is what exposes the invariant line and its actual rank.

3.4. Conjugation. On the conjugate correspondence $\overline{\mathcal{H}(C, \gamma)}$ the endpoint groups are swapped. Its subgroup and base fibre are

$$C^{\text{op}} = \{(h, g) : (g, h) \in C\}, \quad \bar{\gamma}(h, g)\bar{\xi} = \overline{\gamma(g, h)\xi}.$$

The multiplier is $\omega \boxtimes \bar{\mu}$, since $\bar{\gamma}$ conjugates α and swaps its two variables. The label x of a full-germ bundle becomes the inverse germ x^{-1} : indeed

$$(\text{Ad}(h)x \text{Ad}(g^{-1}))^{-1} = \text{Ad}(g)x^{-1} \text{Ad}(h^{-1}).$$

With swapped coset representatives, the transport is explicitly

$$(3.14) \quad U_{(h,g)}^{\bar{X}}(\bar{x}) = \overline{U_{(g,h)}^X(x)}.$$

Thus it includes the conjugate of the scalar ratio in Equation (3.4). At the module level this equality is the defining identity $v_h \bar{\xi} u_g^* = \overline{u_g \xi v_h^*}$. For an original graph $\delta : A \rightarrow B$, the new graph is $\delta^{-1} : B \rightarrow A$ and its fibre representation is $b \mapsto \overline{\sigma(\delta^{-1}(b))}$. Fibre intertwiners are conjugated by the same rule; their adjoints and norms are preserved. Proposition 3.4 also shows directly that the left and right dimensions are exchanged.

3.5. Balanced fusion and its scalar transport. Let $P = L_\nu(\Pi)$. Consider a bundle $\mathcal{H}_X$ from (Γ, μ) to (Λ, ω) and a bundle $\mathcal{H}_Y$ from (Λ, ω) to (Π, ν) , each constructed as above. Write V_x and W_y for their fibres and $\alpha' = \omega \boxtimes \bar{\nu}$. Finite direct sums are handled by distribution, so we may first take one transitive bundle at each endpoint. Define

$$(3.15) \quad Z = (X \times Y)/\Lambda, \quad h(x, y) = ((1, h)x, (h, 1)y).$$

Identify the tensor fibres along this action using

$$(3.16) \quad J_h(x, y) = U_{(1,h)}^X(x) \otimes U_{(h,1)}^Y(y).$$

These identifications are genuine unitary group transports, because their multiplier is $\overline{\omega(h_1, h_2)}\omega(h_1, h_2) = 1$. The action on labels is free, as already its action on X is free. Each quotient fibre V_z is consequently unambiguously a tensor fibre after choosing one representative of z . Put $\mathcal{H}_Z = \bigoplus_{z \in Z} V_z$, with counting norm at any such representatives. Changing representatives gives a unitary identification by Equation (3.16).

The outer transport before passage to the quotient is $U_{(g,1)}^X \otimes U_{(1,k)}^Y$. It commutes exactly with every J_h : each mixed-coordinate factor of either product cocycle is one. Its multiplier is $\mu \boxtimes \bar{\nu}$. For an explicit formula choose a representative (x, y) of z and representatives for all quotient labels. There is a unique h such that

$$((g, h)x, (h, k)y)$$

is the chosen representative of $(g, k)z$. With $s = (g, h)$, $t = (h, k)$, $c = c_X(s, x)$ and $b = c_Y(t, y)$, the quotient transport is

$$(3.17) \quad \frac{\alpha(s, r_x)\alpha'(t, r_y)}{\alpha(r_{sx}, c)\alpha'(r_{ty}, b)}\gamma_X(c) \otimes \gamma_Y(b).$$

It is the product of the two transports in Equation (3.4). There is no additional factor from the compensating element h , since its lift is the genuine transport in Equation (3.16).

To attach a germ to a quotient label, first recall the map from graph cosets to germs. For $x = (g, h)C_\delta \in X$, put

$$d_x = \text{Ad}(h)[\delta] \text{Ad}(g^{-1}),$$

and define d_y in the same way for $y \in Y$. These are the graph-coset-to-germ maps from the earlier regrouping, applied at their respective endpoints; they need not be injective. The middle action changes the two associated germs by

$$(d_x, d_y) \mapsto (\text{Ad}(h)d_x, d_y \text{Ad}(h^{-1})).$$

Hence $z = [(x, y)] \mapsto d_y d_x$ is well defined. Distinct quotient labels can still have the same germ; the next proposition explains how their fibres are regrouped.

Proposition 3.7 (Fusion as a counting tensor-fibre bundle). *There is a unitary correspondence identification*

$$(3.18) \quad \mathcal{H}_X \bar{\otimes}_N \mathcal{H}_Y \cong \mathcal{H}_Z$$

under which outer transport is Equation (3.17). The space Z has finitely many outer-group orbits. For each $z = [(x, y)]$, its stabilizer is a finite-index subgroup-isomorphism graph with germ $d_y d_x$. Regrouping these orbits over full stabilizers by Equation (3.8) gives the full-germ decomposition of the fusion.

Proof. First we check the Hilbert norm and the relative product explicitly. Every $\xi \in V_x$ is right bounded: the map

$$R_\xi : L^2(N) \longrightarrow \mathcal{H}_X, \quad \hat{n} \mapsto \xi n$$

initially defined on the group algebra extends with norm $\|\xi\|$. Indeed the vectors ξv_h occupy distinct fibres as h varies, and all have norm $\|\xi\|$. Choose one x_i in each right Λ -orbit of X , and an orthonormal basis $(e_{ij})_j$ of V_{x_i} . Orthogonality of all their group translates gives

$$R_{e_{ij}}^* R_{e_{i'j'}} = \delta_{ii'} \delta_{jj'} 1.$$

Their ranges exhaust $\mathcal{H}_X$, because the translates exhaust every labelled fibre. They therefore form an orthonormal free right N basis, with finitely many elements.

The defining inner product of the relative tensor product now gives

$$(3.19) \quad \langle e_{ij} \otimes_N \eta, e_{i'j'} \otimes_N \eta' \rangle = \delta_{ii'} \delta_{jj'} \langle \eta, \eta' \rangle.$$

Thus that product is unitarily $\bigoplus_i (V_{x_i} \otimes \mathcal{H}_Y)$. Every label in Z has a unique representative with first coordinate one of the fixed x_i : freeness of the right action on X gives uniqueness of the required balancing element. Consequently the counting tensor-fibre Hilbert sum $\mathcal{H}_Z$ is exactly this same orthogonal sum. Its map to the relative product sends $\xi \otimes \eta \in V_x \otimes W_y$ to $\xi \otimes_N \eta$. Equation (3.19) proves that it is isometric and onto. It is independent of the representative, since

$$(3.20) \quad (\xi v_h^*) \otimes_N (v_h \eta) = \xi \otimes_N (v_h^* v_h \eta) = \xi \otimes_N \eta.$$

Applying the outer generators to these vectors proves Equation (3.17) at the correspondence level.

For finiteness and the stabilizers we give the explicit double-coset calculation, which also specifies the composed germ and its fibre representation. Write the two graph maps as

$$\delta : A \longrightarrow B \leq \Lambda, \quad \varepsilon : C \longrightarrow F \leq \Pi,$$

with $A \leq \Gamma$ and $C \leq \Lambda$ of finite index, and take $X = (\Gamma \times \Lambda)/C_\delta$, $Y = (\Lambda \times \Pi)/C_\varepsilon$. Choose m in the finite double-coset set $B \backslash \Lambda / C$. Represent its outer orbit by

$$z_m = [(C_\delta, (m, 1)C_\varepsilon)].$$

Outer translation and balancing send an arbitrary pair to the form

$$((g, h)C_\delta, (\ell, k)C_\varepsilon) \mapsto (C_\delta, (h^{-1}\ell, 1)C_\varepsilon).$$

Changing the initial coset representatives replaces $h^{-1}\ell$ by $\delta(a)^{-1}h^{-1}\ell c$ for $a \in A$, $c \in C$. Hence these outer orbits are indexed exactly by $B \backslash \Lambda / C$.

An outer element (a, p) stabilizes z_m precisely when some h satisfies

$$(a, h) \in C_\delta, \quad (m^{-1}hm, p) \in C_\varepsilon.$$

Its stabilizer is consequently the graph of

$$(3.21) \quad \begin{aligned} A_m &= \delta^{-1}(B \cap mCm^{-1}), \\ \theta_m &= \varepsilon \circ \text{Ad}(m^{-1}) \circ \delta : A_m \longrightarrow F_m, \\ F_m &= \varepsilon(C \cap m^{-1}Bm). \end{aligned}$$

Both projections have finite index. In particular

$$(3.22) \quad Z \cong \bigsqcup_{m \in B \backslash \Lambda / C} (\Gamma \times \Pi) / C_{\theta_m}.$$

Here C_{θ_m} denotes the graph of the displayed actual map, not necessarily its full-germ stabilizer. At z_m the composite germ is $[\varepsilon \text{Ad}(m^{-1})\delta] = [\theta_m]$. Its graph lies at finite index in its full stabilizer by Lemma 3.1. This proves all the stated regrouping assertions. $\square$

For completeness, the last proof yields an explicit finite double-coset formula without unspecified stabilizer representations. Let σ and τ be the original fibre representations on A and C . In the fibre at z_m choose $r_{(m,1)C_\varepsilon} = (m, 1)$. For $a \in A_m$, write $b = \delta(a)$ and $c = m^{-1}bm \in C$. Equation (3.17), with compensating element $h = b$, gives

$$(3.23) \quad \gamma_m(a) = q_m(b) \sigma(a) \otimes \tau(m^{-1}bm), \quad q_m(b) = \frac{\omega(b, m)}{\omega(m, m^{-1}bm)}.$$

The correction has the exact coboundary

$$(3.24) \quad \frac{q_m(b)q_m(b')}{q_m(bb')} = \frac{\omega(b, b')}{\omega(m^{-1}bm, m^{-1}b'm)}.$$

To verify this, the twisted group-unit identity $v_b v_m = q_m(b) v_m v_{m^{-1}bm}$ can be multiplied successively for b, b' and compared with the identity for bb' . Combining Equation (3.24) with the multipliers of σ and τ shows that γ_m has multiplier $\mu|_{A_m} / \theta_m^* \nu$. Hence

$$(3.25) \quad E(A, B, \delta, \sigma) \bar{\otimes}_N E(C, F, \varepsilon, \tau) \cong \bigoplus_{m \in B \backslash \Lambda / C} E(A_m, F_m, \theta_m, \gamma_m).$$

The appropriate endpoint algebras in the second model and in each summand are understood from their displayed groups. One then applies finite projective induction to the full stabilizers and combines any equal full-germ orbits. Formula (3.23) exhibits explicitly why replacing the twisted fusion rule by its ordinary version would lose a phase.

The indices in Equation (3.21) satisfy

$$[\Gamma : A_m] = [\Gamma : A][B : B \cap mCm^{-1}], \quad [\Pi : F_m] = [\Pi : F][C : C \cap m^{-1}Bm].$$

Counting the left B -orbits in Λ/C and the right C -orbits in $B \backslash \Lambda$ gives, respectively,

$$\sum_m [B : B \cap mCm^{-1}] = [\Lambda : C], \quad \sum_m [C : C \cap m^{-1}Bm] = [\Lambda : B].$$

Thus both coordinate groups have finitely many orbits in the fused label space. These formulas also verify, directly from the models and their counting norms, the usual multiplicativity of the left dimensions and of the right dimensions under this fusion.

3.6. Maps, associativity, and completion of the calculus. The preceding identifications respect the entire operator calculus, not only the isomorphism classes of objects. For the map calculation, first regroup the inputs into full-germ bundles. Theorem 3.5 then represents every bounded intertwiner by an equivariant family of fibre maps. An equivariant family $T_x : V_x \rightarrow V'_x$ and an equivariant family $S_y : W_y \rightarrow W'_y$ give the fibre maps $T_x \otimes S_y$. They commute with the middle identifications in Equation (3.16), so descend to the quotient fibres. Under the unitary of Proposition 3.7, they are precisely $T \bar{\otimes}_N S$, as is seen on $\xi \otimes_N \eta$. Their norms are bounded by $\|T\|\|S\|$; adjoints and compositions agree fibrewise. Regrouping by finite projective induction is a regrouping of these same Hilbert fibres and maps. Conjugation has the map formula already given in Equation (3.14).

For three bundles, both orders of fusion use the quotient of the triple label space by the two middle groups. Their actions commute: on the shared middle bundle they are its commuting coordinate actions, and their projective factors cancel with those in the adjacent bundles. The fibre identification between the two orders is the ordinary unitary

$$(\xi \otimes \eta) \otimes \zeta \mapsto \xi \otimes (\eta \otimes \zeta).$$

The counting norms and Equation (3.19) show that it is also the canonical associativity unitary for the relative products. All scalar identifications are those of Equation (3.16); hence this assertion requires no change of cocycle. For any further number of factors these maps agree on elementary tensors, which are total, so the associativity identifications are coherent. The unit correspondence $L^2(M)$ is the identity-germ model with its one-dimensional trivial graph representation; its unit maps are ordinary left and right multiplication on the corresponding relative tensors.

Conjugation reverses a balanced product. The unitary

$$\overline{\mathcal{H}_X \bar{\otimes}_N \mathcal{H}_Y} \longrightarrow \overline{\mathcal{H}_Y} \bar{\otimes}_N \overline{\mathcal{H}_X}, \quad \overline{\xi \otimes_N \eta} \mapsto \bar{\eta} \otimes_N \bar{\xi}$$

is well defined on tensors of vectors in labelled fibres: the simultaneous middle transport becomes the same transport on the swapped conjugate fibres, with both scalar factors conjugated and their product still one. The counting tensor-fibre model of Proposition 3.7 proves preservation of the norm: the map bijects the quotient labels and is unitary on each conjugate tensor fibre. These tensors are total. It intertwines the swapped endpoint actions and the conjugated fibre maps. Thus the conjugation, unit, and associativity identifications agree with those of the correspondence category at the level of actual operators.

We now apply the geometric input. Proposition 2.1 places every bifinite correspondence with source in $\mathcal{K}$ inside a finite sum of graph models. The preceding full-germ calculation resolves its selecting projection into fibre projections. Together with the operation formulas just proved, this establishes every assertion of Theorem 1.4, including all bounded intertwiners, actual summands and their dimensions, and the specified conjugation and fusion maps. Throughout the section only the matched-subgroup ratio is required to have a finite-dimensional projective realization. None of the calculations imposes a finite-image condition on that representation or a finite-type condition on either individual endpoint cocycle.

4. ARBITRARY FACTOR ENDPOINTS AND STABLE CANONICAL RECOVERY

We now use the correspondence classification to recover the multiplication of an unknown factor. The argument first realizes that factor, up to amplification, as a finite-index extension of the given twisted group factor. Its self-correspondence grading then consists of bounded operators and respects the actual product. This additional multiplication step is what turns the correspondence data into a group presentation. The homogeneous-product and minimal-corner construction has a direct predecessor in Donvil–Vaes [12, Proposition 6.5 and Theorem 6.6]. Their construction already permits arbitrary source cocycles and recovers the exact matched ratio. We adapt it from globally defined automorphism degrees to finite-index isomorphism germs.

4.1. A finite-index extension and its intrinsic grading.

Lemma 4.1 (From a bifinite neighbor to an extension). *Let M, P be II_1 factors and let ${}_M K_P \neq 0$ be bifinite. The commutant Q of the right P -action is a positive finite amplification of P , and the given left action identifies M with a unital finite-index subfactor of Q .*

Proof. The finite right module has a column realization $K = {}_p L^2(P)^n$, with $r = (\text{Tr}_n \otimes \tau_P)(p) = \dim K_P > 0$. Its right-action commutant is $Q = {}_p(M_n(\mathbb{C}) \bar{\otimes} P)p = P^r$, equipped with the normalized corner trace. The left action of M is a normal unital embedding into Q , because a nonzero representation of a factor has zero kernel. Its restriction preserves normalized trace.

As a left Q -module, K has finite positive dimension $1/r$. Choose an integer j with $j/r \geq 1$. Comparison of finite module projections embeds $L^2(Q)$ as a left Q -submodule of $K^{\oplus j}$. Restriction to M makes it a finite left M -module. Thus the Jones index $[Q : M] = \dim_M L^2(Q)$ is finite. This uses the standard finite-factor column and index identifications; see [16] and [17, Theorem 10.1.1, Proposition 10.1.3, and Theorem 10.2.1]. $\square$

Fix $\Gamma \in \mathcal{K}$, an arbitrary normalized scalar cocycle μ , and $M = L_\mu(\Gamma)$. Write $D = \text{Comm}(\Gamma)$ for the group of virtual automorphism germs defined in Section 3. The map

$$\iota : \Gamma \longrightarrow D, \quad \iota(g) = [\text{Ad}(g)],$$

is injective: an element in its kernel centralizes a finite-index subgroup of Γ , and hence has a finite conjugacy class. For a representative $\delta : A \rightarrow B$ of $d \in D$, one has

$$(4.1) \quad d \iota(a) d^{-1} = \iota(\delta(a)) \quad (a \in A).$$

Lemma 4.2 (The intrinsic self-correspondence grading). *Every bifinite M - M correspondence has an orthogonal decomposition*

$$K = \bigoplus_{d \in D} K_d$$

into finite-dimensional spaces, with only countably many nonzero terms, such that

$$u_g K_d u_h^* = K_{\iota(g) d \iota(h)^{-1}}.$$

The support is contained in finitely many $\iota(\Gamma)$ -double cosets. For every representative $\delta : A \rightarrow B$ of d ,

$$(4.2) \quad K_d = \left\{ \xi \in K : \dim_{\mathbb{C}} \text{span}\{u_{\delta(a)} \xi u_a^* : a \in A\} < \infty \right\}.$$

In particular, for the fixed presentation $M = L_\mu(\Gamma)$, the grading is determined by the two actions and does not depend on a choice of arithmetic model decomposition.

Proof. Apply Theorem 1.4 and the full-germ decomposition of Section 3. Use the inverse of its germ labels, so that our degree is a germ from the right group to the left group. If x is an old label and $d = x^{-1}$, the old action gives

$$(\text{Ad}(h)x \text{Ad}(g^{-1}))^{-1} = \iota(g) d \iota(h)^{-1}.$$

Combining the finitely many representation fibres at each identical inverse label gives K_d and the stated covariance. The finitely many old germ orbits become finitely many $\iota(\Gamma)$ -double cosets. Separability makes the nonzero support countable.

For a representative δ , the operators $U_a \xi = u_{\delta(a)} \xi u_a^*$ preserve K_d . Their multiplier is $\mu(\delta(a), \delta(b)) \overline{\mu(a, b)}$. They form a projective unitary representation, so its orbit span is invariant, even when the multiplier is not trivial. This proves the forward inclusion in Equation (4.2).

Conversely let W be a finite-dimensional invariant orbit span and choose an orthonormal basis $\xi_1, \dots, \xi_l$ of W . If P_x projects onto K_x , the numbers

$$m(x) = \sum_{i=1}^l \|P_x \xi_i\|^2$$

have finite sum l and are invariant under $x \mapsto \iota(\delta(a))x\iota(a)^{-1}$. Indeed U_a permutes the degree spaces and carries the chosen basis of W to another orthonormal basis. Positive mass is therefore possible only on a finite orbit. The stabilizer of such an x contains a finite-index subgroup $A_0 \leq A$, and

$$x\iota(a)x^{-1} = \iota(\delta(a)) \quad (a \in A_0).$$

Represent x by $\epsilon : C \rightarrow E$ and restrict to $A_0 \cap C$. Equation (4.1) and injectivity of ι give $\epsilon(a) = \delta(a)$ there. Thus $x = d$. All of W lies in K_d , proving the converse and the asserted intrinsic description. $\square$

4.2. Bounded homogeneous operators and the minimal corner. Let $M \subset Q$ now be a unital finite-index inclusion of II_1 factors and apply Lemma 4.2 to ${}_M L^2(Q)_M$. Until the next lemma, its degree spaces are only Hilbert subspaces of $L^2(Q)$.

Lemma 4.3 (Boundedness and multiplication). *Every K_d consists of bounded elements of Q , and*

$$(4.3) \quad K_d K_f \subset K_{df}, \quad K_d^* = K_{d^{-1}}, \quad 1 \in K_1, \quad u_a \in K_{\iota(a)}.$$

Consequently K_1 is a finite-dimensional unital $$ -subalgebra of Q , normalized by the u_a , and $\tau_Q(K_d) = 0$ for $d \neq 1$.*

Proof. Let $\xi \in K_d$. For distinct $g \in \Gamma$, the right translates ξu_g have distinct degrees $du(g)$. Injectivity of ι therefore gives $\langle \xi, \xi u_g \rangle = 0$ for $g \neq e$. Initially put $a = \xi^* \xi \in L^1(Q)_+$. The L^1 extension of the trace expectation satisfies

$$E_M(a) = \|\xi\|_2^2 1 :$$

indeed, for each $g \in \Gamma$,

$$\tau_M(E_M(a)u_g) = \tau_Q(au_g) = \langle \xi, \xi u_g \rangle.$$

These values agree with those of $\|\xi\|_2^2 1$. They determine the normal functional associated to $E_M(a) \in L^1(M)$, because the group algebra is ultraweakly dense in M . For the bounded spectral truncations $a_n = \min(a, n)$, Pimsner–Popa’s finite-index inequality [23, Proposition 2.1] gives

$$[Q : M]^{-1} a_n \leq E_M(a_n) \leq E_M(a) = \|\xi\|_2^2 1.$$

Monotone convergence makes a bounded, and hence $\xi \in Q$, with $\|\xi\|_\infty \leq [Q : M]^{1/2} \|\xi\|_2$. This argument uses orthogonality of the degrees, so the scalar phases in the right action do not change it. We now verify the product itself.

Take $\xi \in K_d$, $\eta \in K_f$, and representatives $\delta : A \rightarrow B$, $\epsilon : C \rightarrow E$. On the finite-index domain $C_0 = \{a \in C : \epsilon(a) \in A\}$, the exact identity

$$u_{\delta\epsilon(a)} \xi \eta u_a^* = (u_{\delta\epsilon(a)} \xi u_{\epsilon(a)}^*) (u_{\epsilon(a)} \eta u_a^*)$$

puts the orbit of the product into the finite-dimensional span of products of bases of K_d and K_f . These are bounded elements, as just proved. The intrinsic test (4.2) yields $\xi \eta \in K_{df}$. Taking adjoints in the same test gives $K_d^* \subset K_{d^{-1}}$, and taking adjoints again gives equality. The vector 1 is fixed by the identity graph. For the graph of $\text{Ad}(a)$, the vector u_a is carried to a scalar multiple of itself, since the corresponding product of group labels is a . Thus its graph orbit span is one-dimensional, even with the twist. This proves Equation (4.3).

Finite dimension of K_1 comes from the grading lemma, and normalization follows from its covariance. Orthogonality to $1 \in K_1$ gives the trace assertion for other degrees. $\square$

Compressing K_1 to a scalar algebra will now force each surviving degree to be a unitary line.

Lemma 4.4 (The minimal identity corner). *Let p be a minimal projection of K_1 . There are a countable subgroup $\Lambda \leq D$ and a normalized scalar cocycle ω such that the actual corner pQp , with normalized trace, is isomorphic to $L_\omega(\Lambda)$. Each nonzero space $pK_d p$ is one-dimensional and has a unitary basis vector in the corner.*

Proof. Left and right multiplication by p are commuting orthogonal projections on $L^2(Q)$. They preserve each degree by Equation (4.3); their product projects onto $L^2(pQp)$. Thus, up to the common normalization factor of the corner trace,

$$L^2(pQp) = \bigoplus_{d \in D} pK_dp, \quad pK_1p = \mathbb{C}p.$$

For $0 \neq x \in pK_dp$, the product and adjoint laws give $x^*x = \alpha p$ and $xx^* = \beta p$, with positive scalars. Traciality gives $\alpha = \beta$. Therefore $x/\sqrt{\alpha}$ is a unitary in pQp . If y belongs to the same degree, $x^*y \in \mathbb{C}p$, so y is a scalar multiple of x .

Set $\Lambda = \{d : pK_dp \neq 0\}$. It contains the identity, is closed under inverses, and is closed under products because products of its homogeneous unitaries cannot vanish. It is countable by orthogonality and separability. Choose unitary vectors $v_d \in pK_dp$, with $v_1 = p$, and define

$$v_d v_f = \omega(d, f) v_{df}.$$

Unitarity makes the coefficient unimodular, associativity gives the 2-cocycle identity, and $v_1 = p$ gives normalization. The v_d form an orthonormal basis for the normalized corner trace. The unitary $\delta_d \mapsto v_d$ from $\ell^2(\Lambda)$ to $L^2(pQp)$ intertwines twisted left regular multiplication with left multiplication by these units.

Let Q_0 be the von Neumann algebra they generate. The trace-preserving expectation onto Q_0 has L^2 -range containing every v_d , so it has full range and $Q_0 = pQp$. The basis unitary therefore gives a normal unital complex-linear $*$ -isomorphism $L_\omega(\Lambda) \cong pQp$, as claimed. $\square$

4.3. The ratio condition and the reconstructed group. The minimal corner has supplied a twisted group algebra, but the group Λ still needs to satisfy the hypotheses of Theorem 1.3. We next recover a finite-index copy of a subgroup of Γ inside Λ , obtain the projective ratio from the identity matrix block, and prove that Λ is ICC.

Proposition 4.5 (Arithmetic constraints on the corner). *For the corner in Lemma 4.4, the group Λ is countably infinite and ICC. There are finite-index subgroups $A \leq \Gamma$, $B \leq \Lambda$, an actual group isomorphism $\delta : A \rightarrow B$, and a finite-dimensional projective representation with multiplier $\mu|_A / \delta^*(\omega|_B)$.*

Proof. Let z be the central support of p in the finite-dimensional algebra K_1 , so that zK_1 is the matrix block containing p . We first split the action on this block from the homogeneous unitaries that will survive in pQp . The subgroup

$$A = \{a \in \Gamma : u_a z u_a^* = z\}$$

has finite index, because conjugation permutes the finitely many central blocks. Although these conjugations preserve z , they may move p inside its matrix block. We therefore remove their action on that block before compressing. Choose unitary matrix implementers $s_a \in \mathcal{U}(zK_1)$, $s_e = z$, for $\text{Ad}(u_a)|_{zK_1}$. Since these automorphisms form an ordinary group action, there is a normalized scalar cocycle κ with

$$s_a s_b = \kappa(a, b) s_{ab}.$$

These matrices already constitute a finite-dimensional projective representation. Their image is allowed to be infinite.

The unitaries $q_a = s_a^* u_a z \in zQz$ commute with zK_1 . Using $u_a s_b^* u_a^* = s_a s_b^* s_a^*$, their product is

$$\begin{aligned} q_a q_b &= s_a^* (u_a s_b^* u_a^*) u_a u_b z \\ &= \mu(a, b) s_b^* s_a^* u_{ab} z = \mu(a, b) \overline{\kappa(a, b)} q_{ab}. \end{aligned}$$

They are homogeneous of degree $\iota(a)$, and they commute with p . Their compressions $t_a = p q_a p$ are therefore corner unitaries in $pK_{\iota(a)}p$, satisfying the same product formula. It follows that $B = \iota(A)$ lies in Λ , and that $\delta = \iota|_A : A \rightarrow B$ is an actual isomorphism.

To compare their multiplier with the cocycle already chosen in Lemma 4.4, write the basis unitary of degree $\iota(a)$ as $v_{\iota(a)} = \lambda(a)t_a$, where $\lambda : A \rightarrow \mathbb{T}$ and $\lambda(e) = 1$. With the convention

$$(\partial\lambda)(a, b) = \lambda(a)\lambda(b)\lambda(ab)^{-1},$$

we obtain

$$(4.4) \quad \delta^*\omega = (\partial\lambda)\mu|_A \bar{\kappa}, \quad \frac{\mu|_A}{\delta^*\omega} = (\partial\lambda)^{-1}\kappa.$$

On the defining space of the block, the matrices $a \mapsto \overline{\lambda(a)}s_a$ realize precisely the second multiplier. This is the required ratio condition.

We have the subgroup isomorphism and its exact ratio. It remains to show that its image has finite index in Λ and that Λ is ICC. Put $H = \iota(\Gamma)$. Every $d \in D$ commensurates H , by Equation (4.1) on a finite-index domain. Consequently each double coset HdH is a finite union of right H -cosets. Lemma 4.2 puts the original degree support in finitely many such double cosets. Since B has finite index in H , that support is contained in finitely many right B -cosets. The subgroup Λ is contained in this support and contains B ; hence $[\Lambda : B] < \infty$. In particular it is infinite.

Suppose $l \in \Lambda$ has finite conjugacy class. Its centralizer $C_\Lambda(l)$ has finite index in Λ . Thus $A_0 = \iota^{-1}(C_\Lambda(l) \cap B)$ has finite index in Γ , and l centralizes $\iota(A_0)$. Represent its germ by $\epsilon : C \rightarrow E$. For $a \in A_0 \cap C$, centralization and Equation (4.1) give $\iota(\epsilon(a)) = \iota(a)$. Injectivity of ι gives $\epsilon(a) = a$ on this finite-index subgroup. Thus $l = 1$ in D , proving ICC. The construction uses only subgroup isomorphisms and introduces no finite kernel. $\square$

Proof of Theorem 1.3. Apply Lemma 4.1 to the given nonzero bifinite correspondence. It gives a finite-index extension $M \subset Q = P^r$. The preceding lemmas and Proposition 4.5 give $pQp \cong L_\omega(\Lambda)$ with the asserted group and ratio data. Since pQp is the amplification $P^{r\tau_Q(p)}$, standard column equivalences give

$$P \cong L_\omega(\Lambda)^t, \quad t = (r\tau_Q(p))^{-1} > 0.$$

Conversely the stated subgroup and projective data give a nonzero bifinite arithmetic correspondence from M to $L_\omega(\Lambda)$, by Section 3. Compose it with the corner equivalence from that factor to its amplification and then with the specified factor isomorphism to P . This is a nonzero bifinite M - P correspondence.

When $\mu = 1$, conjugating the finite-dimensional ratio representation and transporting it through δ gives a representation with multiplier $\omega|_B$. Projective induction from B to Λ , with the given global multiplier ω , is finite-dimensional and realizes ω itself; compare [12, Lemmas 6.7–6.8]. The induction formula of Section 3 applies equally to this finite-index subgroup. For general μ the construction proves only the exact ratio in Equation (4.4), as required. $\square$

4.4. Stable scales and the actual isomorphism. For a specified stable isomorphism, the category gives a different constraint: its equivalence correspondence has reciprocal dimensions. After integrality forces both dimensions to be one, we will use the given corner columns to identify the specified map. All dimensions below use the normalized factor traces and the ordinary unnormalized matrix trace Tr_n , with $\text{Tr}_n(1) = n$.

Lemma 4.6 (Dimension and corner equivalences). *For composable bifinite correspondences of finite factors, left dimensions multiply under relative tensor product, and so do right dimensions. If $M^t = p(M_n(\mathbb{C}) \bar{\otimes} M)p$ and $C_p = pL^2(M)^n$, then $M^t C_p M$ is an equivalence correspondence with dimensions $(1/t, t)$ in left/right order.*

Proof. For the right-dimension assertion, realize a right N -module H as $eL^2(N)^k$. Then $H \otimes_N K = eK^k$. The normalized trace on the finite factor $\text{End}(K_P)$ restricts to the normalized trace on its unital copy of N . Computing the dimension of eK^k therefore gives

$$\dim(H \otimes_N K)_P = (\text{Tr}_k \otimes \tau_N)(e) \dim K_P.$$

The left formula follows by conjugating the correspondences. The column C_p has right dimension t by its defining projection. The standard column pairings identify $C_p \otimes_M \overline{C_p}$ with $L^2(M^t)$ and $\overline{C_p} \otimes_{M^t} C_p$ with $L^2(M)$. The multiplication formula makes its left dimension $1/t$. These identifications use the normalized corner trace, which accounts for the reciprocal scale. $\square$

Proof of Theorem 1.2. Set $M = L_\omega(\Lambda)$ and $N = L_\nu(\Pi)$. Use the actual given corner presentations

$$M^t = pM_n(M)p, \quad N^s = qM_m(N)q,$$

and the columns $C_p = pL^2(M)^n$, $C_q = qL^2(N)^m$. A specified isomorphism $\theta : M^t \rightarrow N^s$ gives the M – N equivalence correspondence

$$(4.5) \quad H_\theta = \overline{C_p} \otimes_{M^t} \theta L^2(N^s) \otimes_{N^s} C_q.$$

By Lemma 4.6, its left and right dimensions are t/s and s/t . Theorem 1.4 applies to H_θ because $\Lambda \in \mathcal{K}$ and Π is countably infinite ICC. Its full-germ decomposition and Equation (3.13) make both numbers positive integers. Their product is one, so $t = s$.

The same formula now forces a single full-germ orbit, a one-dimensional fibre, and both stabilizer projections equal to the whole endpoint groups. Indeed each nonzero orbit contributes its positive integer fibre dimension times a positive integer subgroup index to each of the two sums, both of which equal one. Thus its graph is an actual group isomorphism $\delta : \Lambda \rightarrow \Pi$. Consequently $\Pi \in \mathcal{K}$, since $\Lambda \in \mathcal{K}$ and this class is invariant under group isomorphism. The one-dimensional fibre is a normalized cochain $b : \Lambda \rightarrow \mathbb{T}$ with

$$\omega(g, h) = b(g)b(h)b(gh)^{-1}\nu(\delta(g), \delta(h)).$$

The arithmetic bundle identification in Section 3 therefore gives $H_\theta \cong {}_\varphi L^2(N)$, where

$$\varphi(u_g) = b(g)v_{\delta(g)}.$$

Multiplication and the displayed cochain equation define a $*$ -homomorphism on the group algebra. Its trace-preserving basis unitary extends it to a normal unital isomorphism $M \rightarrow N$.

We now return to the specified map in its actual corners. First choose any partial isometry between the corresponding projections; then the correspondence identification will supply the inner correction that makes it implement θ . The projections $\varphi^{(n)}(p)$ and q have equal unnormalized trace $t = s$. Comparison of projections in the finite matrix amplification of N gives a rectangular partial isometry

$$z \in M_{m,n}(N), \quad z^*z = \varphi^{(n)}(p), \quad zz^* = q.$$

Define $\theta_0(x) = z\varphi^{(n)}(x)z^*$ on $pM_n(M)p$. Its lifted correspondence in Equation (4.5) is ${}_\varphi L^2(N)$. To check this directly, on bounded columns set $F(\xi) = z\varphi^{(n)}(\xi)$, applying φ entrywise. For $x \in pM_n(M)p$, $a \in M$, and $\xi, \eta \in pM^n$, one has

$$F(x\xi a) = \theta_0(x)F(\xi)\varphi(a), \quad F(\xi)^*F(\eta) = \varphi(\xi^*\eta).$$

Moreover F is onto the bounded columns qN^m , since $zz^* = q$. These identities, followed by the normalized column pairings of Lemma 4.6, contract the two column factors to the trace module with left action φ .

Consequently $H_\theta \cong H_{\theta_0}$. Tensor on the left with C_p and on the right with $\overline{C_q}$; the column equivalences cancel and give a unitary

$$U : {}_\theta L^2(N^s) \longrightarrow {}_{\theta_0} L^2(N^s).$$

A right-linear unitary on this standard trace module is left multiplication by a unitary $w \in N^s$. Its left intertwining identity says $w\theta(x) = \theta_0(x)w$. Replacing z by w^*z , which has the same initial and final projections, therefore yields the actual formula

$$(4.6) \quad \theta(x) = z\varphi^{(n)}(x)z^* \quad (x \in pM_n(M)p).$$

At ordinary unit scale this is precisely $\theta(u_g) = zb(g)v_{\delta(g)}z^*$ with $z \in N$ unitary.

Conversely a group isomorphism and normalized cochain satisfying the cocycle equation construct φ by the same basis unitary. At equal scales, comparison of the two actual projections supplies z , and Equation (4.6) constructs the corner isomorphism. This proves both directions and the specified-map assertion. Taking identical factor inputs and scales 1, t shows that every factor in the statement has fundamental group $\{1\}$. $\square$

5. COMMENSURABILITY, NAMED LATTICES, AND ENDOMORPHISMS

The category calculation gives concise consequences with different mathematical content. Untwisted correspondence existence detects abstract commensurability. Stable isomorphism recovers an entire group and a specified map. A finite-index endomorphism requires one additional classical covolume argument; its two module dimensions are not initially reciprocal.

5.1. Untwisted correspondences and finite-type neighbors.

Corollary 5.1. *Let $\Gamma \in \mathcal{K}$ and let Λ be any countably infinite ICC group. There is a nonzero bifinite $L(\Gamma)$ – $L(\Lambda)$ correspondence if and only if Γ and Λ are abstractly commensurable.*

Proof. A nonzero correspondence has a nonzero full-germ orbit in Theorem 1.4. Its stabilizer is the graph of an actual isomorphism between finite-index subgroups. Conversely such an isomorphism, with its one-dimensional trivial fibre representation, gives the model in Equation (1.3). $\square$

With the finite-type convention of Section 1, the endpoint theorem specializes as follows.

Corollary 5.2. *For $\Gamma \in \mathcal{K}$, a separable II_1 factor P is joined to $L(\Gamma)$ by a nonzero bifinite correspondence if and only if $P \cong L_\omega(\Lambda)^t$ for $t > 0$, an ICC group Λ abstractly commensurable with Γ , and a finite-type cocycle ω on Λ .*

Proof. Theorem 1.3 and the final projective-induction argument in its proof give necessity. For sufficiency, restrict a projective representation of multiplier ω to the matched subgroup and conjugate it; its multiplier is exactly $1/\delta^*\omega$. The graph model and the amplification equivalence give the required correspondence. $\square$

For arbitrary source twist this specialization must be replaced by the ratio in Theorem 1.3. In particular, its direct proof admits the identity correspondence of $L_\mu(\Gamma)$ whatever the individual cocycle μ may be.

5.2. Examples and specified canonical maps. The familiar arithmetic examples are $\text{SL}_3(\mathbb{Z})$, its principal congruence subgroup

$$\Gamma(3) = \ker(\text{SL}_3(\mathbb{Z}) \longrightarrow \text{SL}_3(\mathbb{Z}/3\mathbb{Z})),$$

all their finite-index subgroups, $\text{PSL}_n(\mathbb{Z})$ for $n \geq 4$, and $\text{PSp}_4(\mathbb{Z})$, with the latter two symbols denoting the quotient by the center of the integral matrix group. The arithmetic lattice theorem places these projective groups in the corresponding center-free real groups [5]; the finite-covolume statement appears also in [4, Theorem 1(iv)]. For even n , $\text{PSL}_n(\mathbb{R})$ is the identity root-generated subgroup of $\text{PGL}_n(\mathbb{R})$; for odd n it is all of it. Likewise $\text{PSp}_4(\mathbb{R})$ is the root-generated adjoint symplectic group. These ambient groups have real rank at least two and property (T) [2].

These projective lattices are ICC. A finite conjugacy class gives a finite-index centralizer. That subgroup is Zariski dense by Borel density, so the element centralizes the connected ambient algebraic group and is trivial in its adjoint form [3]. Equivalently, the centralizer formulation in [13, Corollary 3] applies directly to the finite-index lattice and gives the ambient center. For $\text{SL}_3(\mathbb{Z})$ the real center is already trivial. A finite-index subgroup of an ICC group remains ICC: its finite conjugacy class would yield a finite class in the ambient group after using finitely many coset representatives. Thus every group just listed belongs to $\mathcal{K}$.

For any such group Γ , any countably infinite ICC group Λ , and any specified isomorphism

$$\theta : L(\Gamma) \longrightarrow pM_k(L(\Lambda))p,$$

Theorem 1.2 yields

$$(\mathrm{Tr}_k \otimes \tau)(p) = 1, \quad \theta(u_g) = \chi(g)Wv_{\delta(g)}W^*, \quad W^*W = 1, \quad WW^* = p,$$

where $W \in M_{k,1}(L(\Lambda))$, $\delta : \Gamma \rightarrow \Lambda$ is an actual group isomorphism, and $\chi : \Gamma \rightarrow \mathbb{T}$ is a character. This includes the specified column in the specified corner. The character becomes the cochain of Equation (1.1) when scalar twists are present. In particular all the corresponding twisted factors have fundamental group $\{1\}$.

Definition 1.1 also includes ICC lattices in the permitted finite-place groups, mixed real and finite-place products, and products with quaternionic or Cayley real factors of property (T). Finite products of the preceding discrete examples give reducible cases: a non-identity element has a nonidentity coordinate with infinite conjugacy class. More generally the companion exhaustion theorem already allows arbitrary reducible lattices, nonuniform lattices, and torsion. No repetition of the individual geometric proofs is required for these examples.

Corollary 5.3 (Outer automorphisms). *For $\Gamma \in \mathcal{K}$, there is an isomorphism of abstract groups*

$$\mathrm{Out}(L(\Gamma)) \cong \mathrm{Hom}(\Gamma, \mathbb{T}) \rtimes \mathrm{Out}(\Gamma), \quad [\delta] \cdot \chi = \chi \circ \delta^{-1},$$

where Out denotes automorphisms modulo inner automorphisms.

Proof. Put $T_\chi(u_g) = \chi(g)u_g$ and $U_\delta(u_g) = u_{\delta(g)}$. These maps extend normally by their trace-preserving basis unitaries. Characters are invariant under inner group automorphisms, and $U_\delta T_\psi U_\delta^{-1} = T_{\psi \circ \delta^{-1}}$. Thus $(\chi, [\delta]) \mapsto [T_\chi U_\delta]$ is a well-defined homomorphism for the displayed left action. It is surjective by the untwisted unit-scale case of Theorem 1.2, whose cochain is then a character.

For injectivity, suppose $T_\chi U_\delta = \mathrm{Ad}(w)$, and write $b = \chi \circ \delta$ and $w = \sum_x a_x u_x$ in $L^2(L(\Gamma))$. The relation

$$u_{\delta(g)} w u_g^{-1} = b(g)^{-1} w$$

shows that $|a_x|$ is constant on each orbit of $x \mapsto \delta(g)xg^{-1}$. A nonzero coefficient has a finite orbit by square summability. Its stabilizer H has finite index, and $\delta(h) = xhx^{-1}$ for all $h \in H$. Set $\beta = \mathrm{Ad}(x^{-1}) \circ \delta$ and let N be the normal core of H . Since β fixes N pointwise, $g^{-1}\beta(g)$ centralizes N for every $g \in \Gamma$. The centralizer of a finite-index subgroup is trivial by ICC, as in Lemma 3.1; hence $\delta = \mathrm{Ad}(x)$. Now $v = u_x^* w$ implements the pure character twist b . Its Fourier-coefficient absolute values are constant on conjugacy classes, so ICC makes v scalar. Therefore $b = 1$, whence $\chi = 1$, and $[\delta] = 1$, proving injectivity. $\square$

5.3. Recovery from the reduced group C^* -algebra. Write $C_r^*(\Gamma)$ for the operator-norm closure of the group algebra in the left regular representation.

Corollary 5.4 (Reduced C^* -algebra recovery). *Let $\Gamma \in \mathcal{K}$ and suppose that $C_r^*(\Gamma)$ has a unique tracial state. If Λ is any discrete group and there is a complex-linear $*$ -isomorphism $C_r^*(\Gamma) \cong C_r^*(\Lambda)$, then $\Gamma \cong \Lambda$.*

Proof. Let $\alpha : C_r^*(\Gamma) \rightarrow C_r^*(\Lambda)$ be the given $*$ -isomorphism, and denote the canonical traces by τ_Γ and τ_Λ . The isomorphism is unital, and uniqueness gives $\tau_\Lambda \circ \alpha = \tau_\Gamma$. Consequently α extends isometrically to a unitary between the trace-GNS Hilbert spaces. This unitary intertwines left multiplication, so conjugation by it extends α to a normal isomorphism $L(\Gamma) \cong L(\Lambda)$.

The source reduced algebra is separable. Distinct canonical group unitaries in the target have trace 2-norm distance $\sqrt{2}$, and hence operator-norm distance at least $\sqrt{2}$; thus Λ is countable. Moreover, $L(\Lambda)$ is an infinite-dimensional factor, so Λ is infinite and ICC. Indeed, a nonidentity finite conjugacy class would give a nonscalar central class sum. Theorem 1.2, with both cocycles trivial and both amplification scales equal to one, now gives $\Gamma \cong \Lambda$. $\square$

In particular this applies to $\Gamma = \mathrm{SL}_3(\mathbb{Z})$: its reduced group C^* -algebra has a unique tracial state by [1, discussion following Corollary 5].

5.4. Finite-index endomorphisms of the special linear factor. We state the precise classical fact needed in the final deduction. The finite-index image hypothesis is part of it.

Lemma 5.5. *An injective homomorphism $\alpha : \mathrm{SL}_3(\mathbb{Z}) \rightarrow \mathrm{SL}_3(\mathbb{Z})$ whose image has finite index is surjective.*

Proof. Put $G = \mathrm{SL}_3(\mathbb{R})$ and $\Gamma = \mathrm{SL}_3(\mathbb{Z})$. The group G is connected, center-free, simple, of real rank two, and has no compact factor. Both Γ and $\alpha(\Gamma)$ are irreducible lattices in G . The classical strong-rigidity theorem extends their isomorphism to an analytic automorphism Φ of G ; see [25, Theorem B]. This form of strong rigidity allows nonuniform lattices and torsion; its excluded real rank-one $\mathrm{SL}_2(\mathbb{R})$ case does not occur here. The theorem combines the uniform rigidity of Mostow with the nonuniform higher-rank work of Margulis and Raghunathan and Prasad’s extension, as explained in that reference.

The differential $D = d\Phi_e$ preserves the nondegenerate Killing form B on the Lie algebra: $D^\top B D = B$. Hence $|\det D| = 1$. A left-invariant Haar density is therefore preserved at the identity and, by translation, everywhere. Consequently

$$\mathrm{covol}(\alpha(\Gamma)) = \mathrm{covol}(\Phi(\Gamma)) = \mathrm{covol}(\Gamma).$$

On the other hand the first covolume equals $[\Gamma : \alpha(\Gamma)] \mathrm{covol}(\Gamma)$. The common covolume is positive and finite, so the index equals one. $\square$

Corollary 5.6. *Every normal unital complex-linear $*$ -endomorphism θ of $M = L(\mathrm{SL}_3(\mathbb{Z}))$ with $[M : \theta(M)] < \infty$ is surjective.*

Proof. The endomorphism is injective because its kernel is an ultraweakly closed two-sided ideal in a factor, and it preserves the normalized trace. Equip $H = L^2(M)$ with its usual right action and the left action $x\xi = \theta(x)\xi$. It is bifinite, with

$$\dim H_M = 1, \quad \dim_M H = [M : \theta(M)].$$

Its actual full-germ decomposition and Equation (3.13) show that right dimension one leaves exactly one orbit, a one-dimensional fibre, and $A_d = \mathrm{SL}_3(\mathbb{Z})$. The graph is therefore an isomorphism $\delta : \mathrm{SL}_3(\mathbb{Z}) \rightarrow B_d$, where B_d has finite index in the same group. Lemma 5.5 makes B_d the whole group. The left dimension formula now gives $[M : \theta(M)] = 1$, so $\theta(M) = M$. $\square$

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